

Yue Ma

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Profile

Research Engineer and Physics Ph.D. building cost-aware agentic systems that automate long-horizon experimental loops in machine learning and science. Develops evaluation frameworks and trajectory infrastructure to capture, analyze, and improve scientific decision-making under real compute constraints.

Education & Academic Positions

Postdoctoral Researcher, Halicioğlu Data Science Institute, UC San Diego	2026.3 – Present
Ph.D. in Particle Physics, University of California, San Diego	2020.10 – 2026.2
B.S. in Physics, Nanjing University	2016.8 – 2020.6

Agentic Systems, Evaluation, and Research Infrastructure

Resource-Aware Multi-Agent System for Autonomous Research	2026.2 – Present
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A closed-loop agentic system for proposing, implementing, and evaluating machine-learning experiments under online compute constraints. Sole architect and developer; initially deployed on scientific time-series denoising and now extending to additional domains.

- **Multi-Agent Research Runtime:** Designed a typed 6-agent workflow spanning literature review, hypothesis generation, PyTorch implementation, validation, experimentation, and interpretation, with structured findings and failure evidence propagated across iterations.
- **Online Resource Control:** Built dynamic runtime and GPU-memory calibration, isolated per-candidate measurement, and admission control across trial-, formal-, and campaign-level compute budgets.
- **Reliable Long-Horizon Execution:** Implemented scientific-quality gates, watchdogs, bounded retries, fail-closed budget enforcement, and resumable state; achieved over a $5\times$ denoising improvement during autonomous campaigns on the [TIDMAD benchmark](#).

SciTra: Cost-Aware Scientific-Agent Evaluation and Trajectory Infrastructure	2026.7 – Present
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Founder and lead of a 50+ scientist, 20+ institution collaboration developing scientific-agent evaluations and decision-trajectory infrastructure.

- **Cost-Aware Scientific Evaluation:** Formalized open-ended tasks around expensive iterative decisions, treating experiment cost rather than idea generation as the primary scaling bottleneck; coordinated 30+ task ideas across 11 domains.
- **Scientific Decision Trajectories:** Built a cross-vendor runtime that captures expert-agent interactions, tool use, Git changes, failures, and explicit cost, output, and decision records as typed trajectories of how scientific choices are made and revised.
- **Auditable Research Infrastructure:** Implemented append-only declarations, frozen evidence, integrity manifests, interrupted-run recovery, and deterministic export for reproducible trajectory evaluation.

Machine Learning Research

Adaptive Neural Processes for Extreme Forecasting	2024.11 – 2025.6
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Probabilistic forecasting under distribution shifts and limited observations. [arXiv:2602.04609](#)

- **Model Development:** Conceived and derived AdaCNP, a target-conditioned context-reweighting mechanism that enables few-shot adaptation to unseen extreme regimes without retraining.

- **Experimental Evaluation:** Achieved a 22% reduction in MSE and the best negative log-likelihood among Neural Process baselines across two real-world power-system datasets.

Multi-Fidelity Surrogate Modeling for Scientific Simulation

2024.7 – 2024.10

Learning efficient surrogates for computationally expensive, high-variance detector simulations.

- **Scientific ML System:** Developed a multi-fidelity surrogate combining Gaussian Processes and Neural Processes to emulate high-dimensional rare-event simulations.
- **Compute-Efficient Optimization:** Reduced simulation cost by 96.7% while improving background-suppression efficiency by 66.5%; contributed core algorithms and derivations to the [ICLR 2025 Spotlight paper “RESuM”](#) (acknowledged for significant contribution).

Statistical Inference and Large-Scale Scientific Computing

Research within the [XENONnT Collaboration](#), a 200+ scientist international experiment operating petabyte-scale data systems for dark-matter and solar-neutrino searches.

Probabilistic Modeling and Statistical Inference (Analysis Lead)

2020.10 – Present

Leading a team of ~20 scientists developing the first measurement of solar pp neutrinos at ultra-low momentum transfer.

- **Inference and Calibration:** Developed Bayesian and maximum-likelihood inference frameworks, improving the precision of efficiency estimates by 60% through new calibration strategies and data-selection methods.
- **Experimental Evaluation:** Built Monte Carlo workflows for uncertainty propagation, sensitivity estimation, goodness-of-fit testing, and hypothesis testing, informing analysis design and experimental strategy.

Data Infrastructure and High-Throughput Computing

2024.1 – Present

- **Distributed Data Systems:** Key maintainer of the open-source [strax](#) and [straxen](#) frameworks processing petabyte-scale detector data for 200+ collaborators.
- **Performance Optimization:** Reduced ETL processing latency by 40% through vectorized NumPy and Pandas workflows, Numba acceleration, and memory-efficient data processing.

Technical Skills

Programming & ML:	Python, PyTorch, NumPy, Pandas, SciPy, Scikit-learn, C/C++, Bash, Linux, Git
Agents & Evaluation:	LLM agent orchestration, autonomous experimental loops, resource-aware evaluation, trajectory instrumentation, failure diagnostics, multi-agent systems
Systems & Infrastructure:	Pydantic, FastAPI, pytest, GitHub Actions, SLURM/HPC, ETL pipelines, resource monitoring and performance optimization

Selected Publications

Chenxi Hu, **Yue Ma**, Yifan Wu, Yunhe Hou. (2026). "Resilient Load Forecasting under Climate Change: Adaptive Conditional Neural Processes for Few-Shot Extreme Load Forecasting". [arXiv:2602.04609](#).

Aprile, E. et al. (2023). "First Dark Matter Search with Nuclear Recoils from XENONnT". *Phys. Rev. Lett.* 131.

Aprile, E. et al. (2022). "Search for New Physics in Electronic Recoil Data from XENONnT". *Phys. Rev. Lett.* 129.16.